

Project Title:

SNS + SQS Fanout Architecture using AWS

Project Description:

Implemented an event-driven architecture using AWS SNS and SQS to demonstrate fanout messaging. A single producer publishes messages to an SNS topic, which distributes them to multiple SQS queues. Independent worker services consume and process messages asynchronously, simulating real-world microservices such as email notifications and analytics processing.

Architecture Overview:

Producer (EC2)

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SNS Topic

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SQS-1 SQS-2

Email Analytics

Worker Worker

AWS Services Used:

- Amazon SNS
 - Amazon SQS
 - Amazon EC2
 - IAM
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Step-by-Step Implementation:

1. Created SNS Topic

- Created a **Standard SNS Topic**

- Named: fanout-topic
 - Copied **Topic ARN**
-

2 Created SQS Queues

- Created two Standard queues:
 - email-queue
 - analytics-queue
 - Stored:
 - Queue URL
 - Queue ARN
-

3 Subscribed SQS to SNS

- Subscribed both queues to SNS topic:
 - Protocol: Amazon SQS
 - Endpoint: Queue ARN
 - Verified subscription status as **Confirmed**
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4 Configured SQS Access Policy

- Updated SQS queue policy to allow SNS:

```
{  
  "Effect": "Allow",  
  "Principal": { "Service": "sns.amazonaws.com" },  
  "Action": "sqs:SendMessage",  
  "Resource": "QUEUE_ARN",  
  "Condition": {  
    "ArnEquals": {  
      "aws:SourceArn": "SNS_TOPIC_ARN"  
    }  
  }  
}
```

```
}  
}
```

5 Setup EC2 Environment

- Launched EC2 instance
- Installed Python & boto3:

```
sudo apt update
```

```
sudo apt install python3-pip -y
```

```
pip3 install boto3
```

- Attached IAM Role with:
 - SNS access
 - SQS access
-

6 Developed Producer Script

- Created producer.py using boto3
- Published message to SNS topic:

```
sns.publish(  
    TopicArn=topic_arn,  
    Message="New Order Placed"  
)
```

7 Developed Worker Scripts

Email Worker (email_worker.py)

- Polls email-queue
- Receives messages
- Deletes processed messages

Analytics Worker (analytics_worker.py)

- Polls analytics-queue

- Processes same event independently
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8 Executed and Tested System

- Ran workers in separate terminals
 - Triggered producer script
 - Verified both queues received the same message
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Key Features Demonstrated:

- ✓ Fanout messaging using SNS
 - ✓ Decoupled microservices architecture
 - ✓ Asynchronous message processing
 - ✓ Multi-consumer event distribution
 - ✓ Real-time event handling
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Output Result

- Single message published → delivered to multiple queues
- Both workers processed the same event independently

Example:

Email Service received: New Order Placed

Analytics Service received: New Order Placed

Challenges Faced & Fixes :

- Faced **Invalid Topic ARN error** → fixed by using correct SNS ARN
 - Worker not printing → understood SQS long polling behavior
 - Duplicate outputs → identified multiple message publishing
 - Message format confusion → learned SNS wraps message in JSON
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Real-World Use Case:

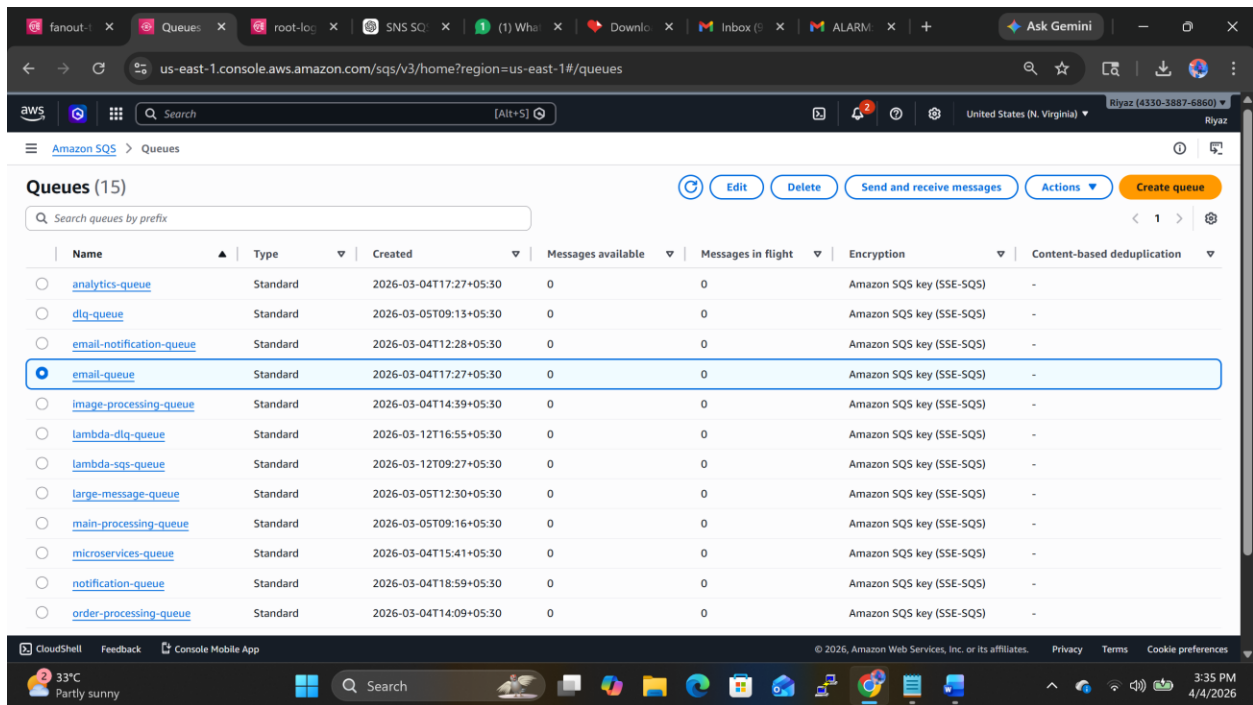
- Order processing system
- One event triggers:
 - Email notification
 - Analytics update
 - Inventory update

SNS topic and subscriptions created:

The screenshot displays the Amazon SNS console interface for a topic named 'fanout-topic'. The left sidebar shows the navigation menu with options like Dashboard, Topics, Subscriptions, and Mobile. The main content area shows the topic details, including its name, ARN, display name, and type (Standard). Below the details, there are tabs for Subscriptions, Access policy, Data protection policy, Delivery policy (HTTP/S), Delivery status logging, Encryption, Tags, and Integrations. The 'Subscriptions' tab is active, showing a list of two subscriptions, both with a status of 'Confirmed' and protocol of 'SQS'. The bottom of the screen shows a Windows taskbar with the date and time as 3:33 PM on 4/4/2026.

ID	Endpoint	Status	Protocol
3117077b-1810-4924-8e5e-2c29ba...	arn:aws:sqs:us-east-1:433038876860:fanout-topic	Confirmed	SQS
77d61a85-fa25-4af4-9900-536997f...	arn:aws:sqs:us-east-1:433038876860:fanout-topic	Confirmed	SQS

SQS queues created(email queue,analytics queue):



Final Output:

